

AssetVal Pro v2 — Model Card & AI Governance Overview

Aligned with ISO 55000, ISO 31000, Responsible AI Principles, Transparency, Governance & Risk Management Practices

1. Model Overview

AssetVal Pro v2 is a decision-support platform designed to support strategic asset management and replacement decisions for infrastructure and industrial assets. The platform combines financial evaluation, technical condition assessment, risk analysis, compliance checks, and multi-criteria decision-making (MCDM) methodologies. The solution supports organizations operating in sectors such as oil & gas, utilities, transport, telecom, and industrial operations.

2. Intended Purpose

The primary purpose of the model is to assist engineers, operations managers, asset managers, and executive decision-makers in evaluating whether an asset should be:

- Retained and maintained
- Upgraded or rehabilitated
- Fully replaced

The model is intended to improve consistency, transparency, and defensibility in capital expenditure and operational decision-making.

3. Functional Components

The platform includes the following core analytical functions:

- Life Cycle Cost (LCC) analysis
- Net Present Value (NPV) calculations
- Risk matrix evaluation (Probability of Failure × Consequence of Failure)
- Technical condition scoring
- Compliance tracking against industry standards
- Multi-Criteria Decision Making (MCDM)
- Sensitivity analysis and comparative scenarios

4. Standards & Regulatory Alignment

The system architecture and methodology align conceptually with internationally recognized frameworks and standards including:

- ISO 55000 — Asset Management
- ISO 31000 — Risk Management
- ASTM E917 — Life Cycle Costing
- API 570 — Piping Inspection Code
- OSHA safety-oriented operational considerations

The governance structure also reflects core Responsible AI and AI governance principles commonly referenced in OECD AI Principles, NIST AI RMF, and emerging AI regulatory frameworks.

5. Transparency & Explainability

The platform is designed to support transparent engineering and financial decisions. Key calculations are visible to the user and based on known engineering and financial formulas.

The system does not operate as a black-box AI model. Users are expected to review:

- Input assumptions
- Risk ratings
- Financial assumptions
- Discount rates
- Asset condition evaluations

Recommendations should always be validated by qualified professionals.

6. Data Requirements

The quality of recommendations depends on the accuracy and completeness of user-provided data. Typical inputs include:

- Installation year
- Design life
- Maintenance costs
- Operating costs
- Asset condition
- Failure risk information
- Compliance status

Incorrect, incomplete, or outdated information may reduce reliability of outputs.

7. Limitations

AssetVal Pro v2 is a decision-support tool and not a replacement for engineering judgment, financial due diligence, or regulatory approval.

The model:

- Does not guarantee operational safety
- Does not replace detailed engineering inspection
- Does not predict failures with certainty
- Depends on user-entered assumptions and parameters

Outputs should be interpreted within the context of organizational strategy, operational conditions, and regulatory requirements.

8. Responsible Use Guidelines

Users should:

- Validate all financial assumptions
- Verify technical inspection data
- Review risk ratings periodically
- Maintain auditability of decisions
- Avoid over-reliance on automated recommendations

The platform is intended to support — not replace — human oversight and accountable decision-making.

9. Privacy & Confidentiality

AssetVal Pro v2 should be deployed in accordance with applicable organizational cybersecurity and data governance requirements.

Recommended practices include:

- Restricting access to authorized personnel
- Protecting operational and financial datasets
- Maintaining audit logs for changes and evaluations
- Applying secure hosting and backup controls

10. Governance & Accountability

Organizations deploying the platform are encouraged to establish:

- Defined approval authority for asset decisions
- Periodic model review processes

- Version control and change management
- Independent validation for major capital decisions

Governance ownership should remain with the deploying organization.

11. Author & Ownership

AssetVal Pro v2

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12. Ethical Statement

The platform is intended to support safer, more sustainable, and economically justified infrastructure management decisions.

The developer encourages responsible deployment practices emphasizing:

- Human oversight
- Operational safety
- Transparency
- Risk awareness
- Professional accountability

Document Purpose: This model card is intended to provide transparency, governance context, intended-use guidance, and operational limitations for AssetVal Pro v2 in line with modern AI and decision-support governance practices.